

Mini Project Title:

"AI-Augmented Live Course Streaming Platform"

Project Objective:

Build a mini system where:

- Instructors can upload **course videos**
 - Students can **stream videos in real-time**, like Netflix
 - AI enhances the experience by:
 - Generating **video summaries**
 - Creating **instant quizzes**
 - Answering doubts via an **AI chatbot**
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Core Functionalities

1. Course Upload (Teacher Side)

- Upload **.mp4** course videos
- Metadata: Title, Topic, Tags, Description
- Store in file system or integrate with dummy CDN-style streaming (simulate via progressive loading or chunking)

2. Course Listing & Playback (Student Side)

- Display list of available courses

- Stream video instantly (simulate using React video player or HLS.js)
- Track watched progress (store in DB)

3. AI Features (Super Value Add)

a. AI-Generated Course Summary

- After upload, extract transcript from video using:
 - **Whisper** (OpenAI) or AssemblyAI
- Send transcript to **OpenAI/Gemini** to generate a summary
- Display below the video as “Quick Recap”

b. Auto Quiz Generator

- From transcript or summary, use AI to create:
 - 2–3 MCQs per video
 - Each with question, options, correct answer, and explanation
- Students can test themselves after each video

Purpose of DB:

- Track uploaded courses
- Store AI-generated summaries
- Keep transcript data

Suggested DB Tables:

- courses
- Transcripts
- Summaries
- Quizzes (optional)

Deliverables:

Github Source Code, API Key, .env file if possible.

Build it and host it if possible.